Confidence for Which Prediction? Supervision Coverage Shapes Grounding Reliability Supplementary Material

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Paper ID *****

A. Two Intervention Blocks

The main paper combines a target-by-readout block and a target-by-positive-coverage block. They address different questions and retain separate fixed recipes and bootstrap draws. Both freeze the localizer and Native selection; confidence never selects a different output box.

Target and readout. For each of MM-GDINO-T and MDETR-R101, the four cells are Global/Selected readout crossed with existence/correct-output labels. Seeds 17/42/73 give 24 heads in this matrix, including the six previously completed MM-GDINO global heads. Positive supervision consists only of parents of the 80,451 negative-text training requests. These are 43,979 unique L1 positives. Full-positive caches and normalization passes must not be mistaken for full-positive supervision in this block.

Target and positive coverage. The additional 12 heads use MM-GDINO and Global-max only: L1-uniform or Full-uniform crossed with E/Y and the same three seeds. The complete L1 pool has 54,015 unique requests. Full adds 25,282 L2 and 4,044 L3 requests, for 83,341 in total. The original paired-L1 heads are scored references, not replacements for the new L1-uniform control.

Every pool is sorted by sample identity before deterministic permutation cycles. A cycle is exhausted before the next begins, and the positive stream continues across the five negative-data epochs. Each head receives 402,255 positive presentations: every L1-uniform example appears 7 or 8 times, every Full-uniform example 4 or 5 times. The total negative presentations remain 402,255, using exactly the original negative order for the matched seed. Within each coverage condition the E/Y positive stream is identical. No additional target-dependent balancing is applied.

These batches contain 32 positives and 32 negatives, not 32 newly created semantic pairs. The loss averages the sources separately, with no paired margin, BatchNorm or cross-example attention. Thus a new positive need not be the original parent of the negative sharing its batch. The negatives retain their true parents in evaluation metadata.

The new control addresses uniform weighting and previously omitted L1 positives before comparing difficulty coverage. Broader coverage still changes the positive images, expressions and relative difficulty weights. The intervention identifies the effect of that specified positive population, not the isolated causal role of an official difficulty label.

B. Localizers and Head Training

MM-GDINO-T. The localizer is the FineCops positive-source epoch-five artifact initialized from released MM-GDINO-T pretraining, not RefCOCO fine-tuning. Its preceding adaptation used all 83,341 positives, one GPU, physical batch 4 with accumulation 8, FP32 AdamW LR 2×10^{-4}, backbone multiplier 0.1, weight decay 10^{-4}, clip 0.1 and epoch-three LR multiplier 0.1. The localizer seed was 20260902; deterministic training was disabled. No-flip multiscale resize/crop used the standard source recipe. Unmatched queries and permitted empty crops still provide background supervision during this earlier adaptation.

Frozen extraction uses full expressions, 800/max1333 resizing, and all 900 final-layer 256-d queries. Native scores are maximum sigmoid token scores; selection preserves the first maximum. The same cached tensors serve every coverage head. Native P@1 on FineCops validation is 77.360%, and every confidence route leaves it unchanged.

MDETR-R101. The second localizer is the official RefCOCO R101 EMA release of MDETR [1]. Only the required EMA weights are used. Official preprocessing produces 100 queries, with Native score $1 - \mathrm{softmax}(\mathrm{logits})_{\text{no object}}$. Selection follows descending score and pixel-xyxy tuple order, with the original index resolving duplicate tuples. Queries are not padded to 900. Correctness uses the selected box's IoU, not an upstream GIoU aggregate. Its FineCops Native P@1 is 65.563%. The coverage intervention is not repeated on this localizer; it tests the scope of the target/readout effect.

Heads and optimization. Detached 256-d query features and the Native scalar form a 257-d input. LayerNorm, Linear(257,128), GELU, Linear(128,128), GELU and Linear(128,1) give 50,179 parameters in eight tensors. The final weight and bias start at zero. All compared heads use the original initialization factory before copying weights to their independent owners. Selected heads are not continued from Global checkpoints; coverage heads are not continued from paired-L1 heads.

For positive logits p_i and negative logits n_i,

$$L = \tfrac{1}{2}\overline{\mathrm{BCE}(p_i, y_i)} + \tfrac{1}{2}\overline{\mathrm{BCE}(n_i, 0)} + \tfrac{10^{-3}}{2}\left(\overline{p_i^2} + \overline{n_i^2}\right). \quad (1)$$

E labels all positive requests one; Y labels them by the fixed Native IoU ≥ 0.5 event. AdamW uses LR 10^{-4}, WD zero, clip 0.1 and deterministic FP32. Cached features are FP16; scores, boxes and head computation are FP32. Each negative epoch contains 2,514 full batches and one batch with three requests per source, giving 2,515 updates and 12,575 at the fixed endpoint. No checkpoint, learning rate, threshold or population is selected by evaluation results. Complete endpoints and their successful process exits precede scoring.

C. Evaluation Populations and Scope

FineCops validation contains 9,426 positives and 9,029 edited-text negatives on 3,567 images. Its positive levels are 6,097 L1, 2,884 L2 and 445 L3; the MM-GDINO correct/wrong counts are 5,506/591, 1,566/1,318 and 220/225. All 9,029 negatives have L1 positive parents, including repeated edits from 4,946 unique parents. Difficulty comparisons use those parents' positive levels, never the edit's separate difficulty annotation.

gRefCOCO TestA has 5,917 single-target positives and 4,448 no-targets; TestB has 5,646 and 4,673. The pooled 20,684 requests span 1,500 images. All 14,579 multi-target expressions are excluded: these results do not evaluate full zero-to-many-target GREC. The FineCops-source-disjoint sensitivity removes 202 training-source and 21 validation-source images through VG–COCO identities or exact bytes. It retains 1,277 images,9,848 positives and 7,716 no-targets. Full and source-disjoint overlap. Neither this exclusion nor frozen transfer claims disjointness from all pretraining, MDETR's RefCOCO adaptation, or unknown near duplicates.

No new FineCops Test or negative-image evaluation is performed. The readout study traverses each localizer once for gRef and reuses its cache. Coverage evaluation reuses the complete MM-GDINO cache without another detector traversal. The old Native boxes, scores, masks and correctness match record by record. No-target correctness and GT IoU are null, not fabricated negative boxes. Validation scoring preserves its original batches of 32; gRef preserves the original four-worker grouping for exact numerical parity.

D. Three-State Risk and Statistical Comparisons

Let $p_C = (1-\pi)a$, $p_W = (1-\pi)(1-a)$ and $p_N = \pi$. Pairwise AUROC gives half credit to ties. Then

$$\mathrm{AUC}_E = aU_{CN} + (1-a)U_{WN}, \quad (2)$$

$$\mathrm{AUGRC} = p_C p_W(1-U_{CW}) + p_C p_N(1-U_{CN}) + \tfrac{1}{2}(p_W + p_N)^2. \quad (3)$$

Taking differences gives the main-paper identity. Its C/W and C/N terms are pairwise contributions to total-risk change, not individually equivalent to the wrong-box and no-target components of a selective-risk curve.

For Full−L1 supervision on FineCops, Y's pairwise contributions are $+0.521$ (C/W) and -0.853 (C/N), summing to-0.332 AUGRC points. E's are $+1.526$ and -0.609, summing to$+0.916$. These are exact point decompositions of existing estimates, not newly bootstrapped component intervals or causal explained-variance shares.

The primary coverage effects are $D_Y = R_{\mathrm{Full},Y} - R_{\mathrm{L1},Y}$, $D_E = R_{\mathrm{Full},E} - R_{\mathrm{L1},E}$ and $I = D_Y - D_E$. The absolute effects and residual Y−E gaps are reported along with the interaction; a negative interaction alone need not improve Y. The same definition with S/G describes the separate readout block.

Each block uses 5,000 paired image-cluster bootstrap draws with PCG64: seed 20260911 for readout, 20260912 for coverage. Within each block and evaluation surface all arms and the three seeds receive the same draw; gRef is stratified by TestA/B. Metrics are recomputed per seed and averaged equally. The historical paired-L1 scores enter the coverage draws as fixed references; their point estimates reproduce the original results, but their intervals use the new draw sequence. Image uncertainty is conditional on those heads; sample SD and individual seed effects are reported separately. There is no $n = 3$ t-test or resampling of trained seeds. Intervals are exploratory, with no deletion of undefined replicates.

FPR95 recalculates positive q05 within every replicate and accepts equality. It is a diagnostic, not a fitted deployment threshold. AURC admits complete tie groups and integrates selective risk by trapezoids, extending the first nonempty group's risk to zero coverage. AUGRC instead starts at zero generalized risk and zero coverage. Both integrals and achieved fixed coverages are retained rather than assumed to select the same winner.

For prevalence curves, positive and no-target conditional distributions are held fixed while their masses become $1-\pi$ and π. A crossover solves $(1-\pi)(1-a)\Delta U_{CW} + \pi\Delta U_{CN} = 0$. The root of the mean effect is not the mean of seed roots. If any draw has no internal root, an unconditional root interval is not reported. A mean curve with no root is not a simultaneous statistical guarantee.

E. Conditioning, Readout Diagnostics and Combinations

Same-image analysis carries all eligible state pairs in an image; bootstrap multiplicity m_i weights those pairs by m_i, notm_i^2. Its unconditional reference uses the same participating requests. Difficulty contributions partition within- and across-level pairs; they sum to the full AUROC but are not causal fractions. Same-level L2/L3 C/N comparisons are undefined because there are no corresponding negative parents. The cross-level contribution instead compares L2/L3 positives with the existing L1-parent negatives.

Under paired-L1 MM-GDINO supervision, the L1 Y$-$E means improve both C/W and C/N, implying no internal AUGRC crossover for the fixed L1 mean population. That is an analytic consequence of existing means, not a new conditional risk interval. The subsequent coverage intervention changes the training population and is distinct from conditioning evaluation records.

The readout block retains both Global and Selected logits at fixed weights, Native/confidence winner indices, winner-box overlap and GT IoUs. A different query index need not mean a different box. For a Selected-trained head, the Global winner is counterfactual, not the deployed readout. Inference-only switching and matched retraining follow different paths and do not define a unique spatial contribution. MDETR's FineCops Global-emission inference change is$+4.726, -0.061, +0.158$ across seeds, versus$-0.349, -0.075, -0.143$ for matched retraining; the large inference mean has sample SD 2.703.

The fixed combinations use the original paired-L1 Global-existence head. Product is$\log\max(s_0, 10^{-6}) + \log\sigma(z_E)$. SIRC-style [2] is

$$\begin{aligned} c_{\text{SIRC}} = & -\log\max(1-s_0, 10^{-6}) \\ & - \operatorname{softplus}\big(-(z_E - (\mu - 3\sigma))/\sigma\big). \end{aligned} \tag{4}$$

Mean and population SD use all 83,341 unique training positives for that localizer/seed; this normalization is not supervised head training. At SD$\leq 10^{-12}$ the rule returns Native. No evaluation weights or thresholds are fitted. These controls are compared with Native and the original learned heads, not claimed to dominate the Full-coverage heads.

F. Coverage Results: Every Seed and Endpoint

The following tables report all new cells, original paired-L1 references, primary effects and secondary metrics. Main-text risk conclusions concern the observed mixtures. Crossover counts and conditional C/N tables retain the residual tradeoffs and empty comparison cells.

Table 1. FineCops val: complete coverage-block endpoints, mean (sample SD), all $\times 100$. Paired-L1 is the reused reference, not the new L1-uniform control. FPR95 is diagnostic; no deployment threshold is fitted.

Score	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Native	28.11 (0.00)	83.28 (0.00)	53.16 (0.00)	18.71 (0.00)	45.36 (0.00)	54.57 (0.00)	96.09 (0.00)
Paired-L1 E	26.23 (0.07)	68.89 (1.99)	66.30 (0.27)	50.05 (2.09)	62.62 (0.65)	47.39 (0.25)	80.11 (0.13)
Paired-L1 Y	26.38 (0.04)	80.55 (0.40)	62.78 (0.12)	32.26 (0.56)	55.87 (0.09)	47.69 (0.17)	85.73 (0.22)
L1-uniform E	25.88 (0.06)	67.08 (4.22)	68.53 (1.29)	53.78 (5.81)	65.19 (2.31)	46.15 (0.13)	80.41 (0.99)
L1-uniform Y	26.02 (0.04)	79.24 (0.63)	64.94 (0.37)	35.03 (1.28)	58.17 (0.58)	46.27 (0.06)	86.11 (0.24)
Full-uniform E	26.80 (0.23)	33.69 (3.08)	71.68 (0.47)	81.14 (2.00)	73.82 (0.11)	52.80 (0.84)	78.03 (0.20)
Full-uniform Y	25.69 (0.02)	67.83 (1.38)	69.35 (0.30)	53.31 (1.82)	65.72 (0.63)	45.82 (0.10)	81.78 (0.55)

Table 2. FineCops val: every new head seed; raw endpoints $\times 100$.

Score	Seed	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
L1-uniform E	17	25.86	63.77	69.43	58.20	66.89	46.17	79.64
L1-uniform E	42	25.95	71.84	67.05	47.20	62.56	46.26	81.53
L1-uniform E	73	25.84	65.63	69.11	55.95	66.13	46.01	80.05
L1-uniform Y	17	26.00	79.05	65.07	35.32	58.33	46.27	86.14
L1-uniform Y	42	26.07	79.94	64.52	33.64	57.53	46.33	86.33
L1-uniform Y	73	25.99	78.74	65.24	36.15	58.65	46.21	85.86
Full-uniform E	17	27.06	30.16	71.14	83.42	73.92	53.77	77.80
Full-uniform E	42	26.65	35.81	71.95	79.71	73.71	52.23	78.10
Full-uniform E	73	26.68	35.10	71.95	80.29	73.84	52.39	78.19
Full-uniform Y	17	25.70	66.30	69.63	55.33	66.39	45.93	81.23
Full-uniform Y	42	25.66	68.21	69.40	52.81	65.64	45.75	81.77
Full-uniform Y	73	25.70	68.97	69.03	51.80	65.13	45.77	82.33

Table 3. FineCops val: coverage/target contrasts with paired 95% intervals, $\times 100$.

Contrast	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Full−L1, Y	-0.332 [-0.436, -0.235]	-11.411 [-12.237, -10.582]	+4.412 [+3.956, +4.885]	+18.281 [+17.300, +19.286]	+7.552 [+7.039, +8.076]	-0.450 [-0.742, -0.174]	-4.334 [-5.151, -3.473]
Full−L1, E	+0.916 [+0.755, +1.078]	-33.392 [-35.041, -31.809]	+3.153 [+2.494, +3.841]	+27.359 [+25.965, +28.812]	+8.633 [+7.895, +9.408]	+6.651 [+5.906, +7.418]	-2.374 [-3.092, -1.537]
Interaction	-1.248 [-1.366, -1.138]	+21.981 [+20.594, +23.469]	+1.259 [+0.846, +1.667]	-9.077 [-10.410, -7.846]	-1.081 [-1.602, -0.594]	-7.101 [-7.782, -6.448]	-1.960 [-3.050, -0.923]
L1: Y−E	+0.138 [+0.053, +0.223]	+12.160 [+11.287, +13.080]	-3.587 [-3.948, -3.239]	-18.749 [-19.747, -17.758]	-7.019 [-7.482, -6.571]	+0.122 [-0.068, +0.309]	+5.704 [+4.869, +6.638]
Full: Y−E	-1.110 [-1.279, -0.951]	+34.141 [+32.425, +35.993]	-2.327 [-2.948, -1.730]	-27.827 [-29.356, -26.372]	-8.100 [-8.824, -7.390]	-6.979 [-7.763, -6.204]	+3.743 [+2.891, +4.567]

Table 4. gRef source-disjoint: complete coverage-block endpoints, mean (sample SD), all $\times 100$. Paired-L1 is the reused reference, not the new L1-uniform control. FPR95 is diagnostic; no deployment threshold is fitted.

Score	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Native	30.05 (0.00)	72.46 (0.00)	68.09 (0.00)	48.65 (0.00)	59.55 (0.00)	55.97 (0.00)	81.36 (0.00)
Paired-L1 E	28.68 (0.08)	66.46 (0.93)	81.41 (0.19)	68.63 (0.80)	75.79 (0.39)	51.18 (0.43)	72.89 (2.03)
Paired-L1 Y	28.61 (0.04)	72.99 (0.46)	78.22 (0.14)	61.43 (0.34)	70.84 (0.14)	50.79 (0.32)	70.12 (0.67)
L1-uniform E	28.65 (0.04)	65.94 (1.04)	81.94 (0.33)	68.66 (0.82)	76.10 (0.55)	51.80 (0.30)	74.15 (1.05)
L1-uniform Y	28.49 (0.04)	72.92 (0.03)	79.16 (0.29)	61.98 (0.24)	71.61 (0.19)	50.49 (0.31)	73.01 (2.03)
Full-uniform E	29.69 (0.12)	55.12 (0.96)	80.45 (0.32)	71.85 (0.58)	76.67 (0.16)	57.80 (0.45)	76.65 (3.87)
Full-uniform Y	28.33 (0.07)	72.27 (0.38)	80.64 (0.33)	64.10 (0.58)	73.37 (0.36)	49.81 (0.36)	72.85 (1.38)

Table 5. gRef source-disjoint: every new head seed; raw endpoints $\times 100$.

Score	Seed	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
L1-uniform E	17	28.64	65.77	82.06	68.90	76.27	51.88	74.29
L1-uniform E	42	28.61	67.04	81.56	67.74	75.49	51.46	75.13
L1-uniform E	73	28.68	64.99	82.20	69.33	76.54	52.05	73.04
L1-uniform Y	17	28.49	72.95	79.10	61.71	71.46	50.52	75.22
L1-uniform Y	42	28.53	72.89	78.90	62.18	71.55	50.79	71.23
L1-uniform Y	73	28.44	72.92	79.47	62.06	71.82	50.17	72.58
Full-uniform E	17	29.82	54.03	80.08	72.44	76.72	58.32	74.42
Full-uniform E	42	29.60	55.83	80.69	71.83	76.79	57.48	74.40
Full-uniform E	73	29.64	55.50	80.58	71.27	76.49	57.60	81.12
Full-uniform Y	17	28.27	72.43	81.01	64.44	73.73	49.55	71.70
Full-uniform Y	42	28.40	71.83	80.39	64.42	73.37	50.22	72.47
Full-uniform Y	73	28.33	72.54	80.53	63.43	73.01	49.66	74.38

Table 6. gRef source-disjoint: coverage/target contrasts with paired 95% intervals, $\times 100$.

Contrast	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Full−L1, Y	-0.154 [-0.223, -0.083]	-0.657 [-1.201, -0.118]	+1.483 [+1.202, +1.757]	+2.115 [+1.766, +2.472]	+1.761 [+1.518, +2.015]	-0.681 [-0.989, -0.374]	-0.160 [-1.519, +1.231]
Full−L1, E	+1.043 [+0.917, +1.169]	-10.814 [-11.955, -9.698]	-1.491 [-1.947, -1.045]	+3.189 [+2.659, +3.743]	+0.566 [+0.187, +0.959]	+6.002 [+5.229, +6.757]	+2.493 [+0.737, +3.625]
Interaction	-1.197 [-1.306, -1.088]	+10.158 [+9.149, +11.150]	+2.974 [+2.595, +3.365]	-1.074 [-1.511, -0.633]	+1.194 [+0.872, +1.518]	-6.683 [-7.370, -5.988]	-2.652 [-4.175, -0.513]
L1: Y−E	-0.157 [-0.259, -0.056]	+6.988 [+6.070, +7.911]	-2.781 [-3.169, -2.385]	-6.676 [-7.288, -6.080]	-4.493 [-4.872, -4.115]	-1.308 [-1.746, -0.883]	-1.145 [-2.527, +0.117]
Full: Y−E	-1.355 [-1.521, -1.187]	+17.145 [+15.589, +18.683]	+0.193 [-0.373, +0.777]	-7.750 [-8.618, -6.918]	-3.298 [-3.831, -2.774]	-7.991 [-8.886, -7.086]	-3.797 [-5.458, -1.643]

Table 7. gRef Full: complete coverage-block endpoints, mean (sample SD), all $\times 100$. Paired-L1 is the reused reference, not the new L1-uniform control. FPR95 is diagnostic; no deployment threshold is fitted.

Score	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Native	30.24 (0.00)	71.79 (0.00)	67.69 (0.00)	48.90 (0.00)	59.41 (0.00)	56.38 (0.00)	81.19 (0.00)
Paired-L1 E	28.79 (0.08)	66.52 (0.95)	81.18 (0.19)	68.46 (0.78)	75.58 (0.37)	51.35 (0.43)	73.13 (2.00)
Paired-L1 Y	28.75 (0.05)	72.71 (0.48)	78.01 (0.14)	61.53 (0.35)	70.75 (0.13)	51.06 (0.33)	70.18 (0.60)
L1-uniform E	28.75 (0.04)	66.11 (1.03)	81.69 (0.31)	68.49 (0.79)	75.87 (0.52)	51.91 (0.28)	74.22 (0.95)
L1-uniform Y	28.62 (0.05)	72.74 (0.08)	78.95 (0.31)	62.03 (0.25)	71.50 (0.18)	50.76 (0.34)	73.03 (2.30)
Full-uniform E	29.78 (0.12)	55.40 (0.98)	80.23 (0.35)	71.70 (0.56)	76.47 (0.17)	57.84 (0.46)	76.33 (4.06)
Full-uniform Y	28.46 (0.07)	72.18 (0.39)	80.40 (0.33)	64.09 (0.58)	73.21 (0.35)	50.05 (0.37)	72.72 (1.24)

Table 8. gRef Full: every new head seed; raw endpoints $\times 100$.

Score	Seed	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
L1-uniform E	17	28.75	65.89	81.79	68.73	76.04	52.02	74.26
L1-uniform E	42	28.71	67.23	81.34	67.61	75.29	51.59	75.15
L1-uniform E	73	28.79	65.20	81.93	69.12	76.29	52.13	73.25
L1-uniform Y	17	28.62	72.77	78.91	61.76	71.36	50.78	75.49
L1-uniform Y	42	28.67	72.65	78.66	62.25	71.43	51.10	70.92
L1-uniform Y	73	28.57	72.80	79.28	62.08	71.71	50.41	72.69
Full-uniform E	17	29.92	54.28	79.84	72.27	76.50	58.37	74.16
Full-uniform E	42	29.68	56.13	80.51	71.70	76.63	57.52	73.82
Full-uniform E	73	29.73	55.79	80.34	71.14	76.29	57.63	81.02
Full-uniform Y	17	28.40	72.33	80.77	64.44	73.57	49.78	71.81
Full-uniform Y	42	28.53	71.74	80.14	64.40	73.20	50.47	72.22
Full-uniform Y	73	28.45	72.47	80.30	63.42	72.87	49.89	74.14

Table 9. gRef Full: coverage/target contrasts with paired 95% intervals, $\times 100$.

Contrast	AUGRC	U_{CW}	U_{CN}	U_{WN}	AUC$_E$	AURC	FPR95
Full−L1, Y	-0.157 [-0.222, -0.091]	-0.559 [-1.085, -0.031]	+1.451 [+1.179, +1.711]	+2.054 [+1.714, +2.383]	+1.717 [+1.474, +1.949]	-0.715 [-0.985, -0.444]	-0.311 [-1.554, +0.974]
Full−L1, E	+1.026 [+0.907, +1.141]	-10.706 [-11.704, -9.668]	-1.457 [-1.882, -1.038]	+3.215 [+2.731, +3.700]	+0.601 [+0.256, +0.954]	+5.923 [+5.219, +6.598]	+2.116 [+0.830, +3.453]
Interaction	-1.183 [-1.281, -1.081]	+10.147 [+9.254, +11.027]	+2.908 [+2.551, +3.269]	-1.161 [-1.562, -0.770]	+1.115 [+0.812, +1.418]	-6.637 [-7.247, -6.018]	-2.427 [-4.146, -0.749]
L1: Y−E	-0.134 [-0.229, -0.039]	+6.634 [+5.841, +7.472]	-2.736 [-3.094, -2.381]	-6.455 [-7.043, -5.887]	-4.374 [-4.739, -4.018]	-1.152 [-1.536, -0.763]	-1.184 [-2.367, +0.120]
Full: Y−E	-1.316 [-1.463, -1.161]	+16.780 [+15.392, +18.166]	+0.172 [-0.358, +0.721]	-7.616 [-8.381, -6.869]	-3.259 [-3.762, -2.763]	-7.789 [-8.578, -6.958]	-3.611 [-5.323, -1.831]

Table 10. FineCops C/N changes: full AUROC, L1-conditional AUROC and weighted within/cross-level contributions. The last two columns sum to the first, not to the conditional column. All N parents are L1; cross-level means L2/L3 C against those N. Conditional L2/L3 same-level C/N is undefined, not zero.

Contrast	Full C/N	L1 C/N	Within contribution	Cross contribution
Full−L1, Y	+4.412 [+3.956, +4.885]	-1.019 [-1.294, -0.746]	-0.770 [-0.976, -0.561]	+5.182 [+4.789, +5.592]
Full−L1, E	+3.153 [+2.494, +3.841]	-3.793 [-4.296, -3.287]	-2.864 [-3.240, -2.481]	+6.017 [+5.544, +6.507]
Interaction	+1.259 [+0.846, +1.667]	+2.774 [+2.418, +3.141]	+2.094 [+1.826, +2.372]	-0.835 [-1.118, -0.564]
L1: Y−E	-3.587 [-3.948, -3.239]	+0.112 [-0.076, +0.304]	+0.085 [-0.057, +0.228]	-3.671 [-4.006, -3.365]
Full: Y−E	-2.327 [-2.948, -1.730]	+2.886 [+2.420, +3.357]	+2.179 [+1.831, +2.531]	-4.506 [-4.936, -4.087]

Table 11. Y−E AUGRC crossover in no-target prior. Class-conditional requests remain fixed. An unconditional root interval is given only when all 5,000 replicates have an interior root; missing roots are not imputed. A mean curve without a root does not imply a uniform confidence guarantee.

Population	Supervision	Mean root	95% CI	Interior draws
FineCops val	L1	0.434	[0.401, 0.467]	5000/5,000
FineCops val	Full	0.769	[0.721, 0.819]	5000/5,000
gRef source-disjoint	L1	0.525	[0.471, 0.580]	5000/5,000
gRef source-disjoint	Full	none	–	1275/5,000
gRef Full	L1	0.516	[0.464, 0.569]	5000/5,000
gRef Full	Full	none	–	1390/5,000

G. Target/Readout and Combination Results

The paired-L1 block below is retained in full, including transfer effects, seed variability, fixed-weight cross-readouts and combination costs. Its Selected/Global contrasts should not be confused with the coverage block's Full/L1 contrasts. Winner geometry is descriptive; the matched four cells use their trained readouts.

Table 12. Fixed combinations on the same Native boxes. Mixed AUGRC $\times 100$, and paired differences against all three prespecified references. Lower is better. Product and SIRC-style use Native and global-existence scores; only SIRC-style uses training-positive score statistics. No evaluation mixture, weight or threshold is fitted.

Localizer	Population	Score	AUGRC (SD)	Δ Native	$\Delta G/E$	$\Delta G/Y$
MM-GDINO-T	FineCops val	Product	26.62 (0.03)	-1.497 [-1.598, -1.398]	+0.386 [+0.286, +0.488]	+0.238 [+0.177, +0.302]
MM-GDINO-T	FineCops val	SIRC-style	27.35 (0.02)	-0.758 [-0.821, -0.696]	+1.125 [+0.972, +1.280]	+0.977 [+0.865, +1.090]
MM-GDINO-T	gRef source-disjoint	Product	28.48 (0.03)	-1.569 [-1.727, -1.407]	-0.196 [-0.282, -0.111]	-0.130 [-0.187, -0.074]
MM-GDINO-T	gRef source-disjoint	SIRC-style	28.71 (0.01)	-1.338 [-1.482, -1.191]	+0.035 [-0.088, +0.157]	+0.101 [+0.040, +0.164]
MDETR-R101	FineCops val	Product	29.33 (0.18)	-0.831 [-0.999, -0.663]	-0.381 [-0.439, -0.321]	+0.485 [+0.385, +0.589]
MDETR-R101	FineCops val	SIRC-style	30.01 (0.00)	-0.146 [-0.159, -0.134]	+0.304 [+0.107, +0.493]	+1.170 [+1.038, +1.302]
MDETR-R101	gRef source-disjoint	Product	19.84 (0.46)	-1.291 [-1.485, -1.088]	-0.265 [-0.295, -0.237]	-0.076 [-0.170, +0.015]
MDETR-R101	gRef source-disjoint	SIRC-style	20.95 (0.03)	-0.184 [-0.204, -0.156]	+0.843 [+0.644, +1.039]	+1.031 [+0.881, +1.179]

Table 13. Mixed AUGRC by seed, $\times 100$. Native repeats one frozen prediction route, not three localizer training seeds. Part 1.

Localizer	Population	Score	Seed 17	Seed 42	Seed 73
MM-GDINO-T	FineCops val	Native	28.11	28.11	28.11
MM-GDINO-T	FineCops val	G/E	26.17	26.31	26.21
MM-GDINO-T	FineCops val	G/Y	26.36	26.42	26.35
MM-GDINO-T	FineCops val	S/E	26.12	26.27	26.20
MM-GDINO-T	FineCops val	S/Y	26.32	26.40	26.35
MM-GDINO-T	FineCops val	Product	26.62	26.64	26.59
MM-GDINO-T	FineCops val	SIRC-style	27.37	27.36	27.34
MDETR-R101	FineCops val	Native	30.16	30.16	30.16
MDETR-R101	FineCops val	G/E	29.65	30.00	29.47
MDETR-R101	FineCops val	G/Y	28.98	28.74	28.79
MDETR-R101	FineCops val	S/E	29.54	29.78	29.23
MDETR-R101	FineCops val	S/Y	28.64	28.67	28.65
MDETR-R101	FineCops val	Product	29.25	29.53	29.20
MDETR-R101	FineCops val	SIRC-style	30.01	30.01	30.01
MM-GDINO-T	gRef source-disjoint	Native	30.05	30.05	30.05
MM-GDINO-T	gRef source-disjoint	G/E	28.61	28.76	28.66
MM-GDINO-T	gRef source-disjoint	G/Y	28.61	28.65	28.57
MM-GDINO-T	gRef source-disjoint	S/E	28.53	28.75	28.67
MM-GDINO-T	gRef source-disjoint	S/Y	28.58	28.67	28.65
MM-GDINO-T	gRef source-disjoint	Product	28.49	28.52	28.45
MM-GDINO-T	gRef source-disjoint	SIRC-style	28.72	28.72	28.70
MDETR-R101	gRef source-disjoint	Native	21.13	21.13	21.13
MDETR-R101	gRef source-disjoint	G/E	20.57	20.23	19.52
MDETR-R101	gRef source-disjoint	G/Y	19.54	20.30	19.91

Table 14. Mixed AUGRC by seed, ×100. Native repeats one frozen prediction route, not three localizer training seeds. Part 2.

Localizer	Population	Score	Seed 17	Seed 42	Seed 73
MDETR-R101	gRef source-disjoint	S/E	20.29	20.51	20.23
MDETR-R101	gRef source-disjoint	S/Y	20.41	20.37	20.28
MDETR-R101	gRef source-disjoint	Product	20.26	19.90	19.36
MDETR-R101	gRef source-disjoint	SIRC-style	20.97	20.92	20.96
MM-GDINO-T	gRef Full	Native	30.24	30.24	30.24
MM-GDINO-T	gRef Full	G/E	28.72	28.88	28.77
MM-GDINO-T	gRef Full	G/Y	28.75	28.80	28.70
MM-GDINO-T	gRef Full	S/E	28.65	28.87	28.78
MM-GDINO-T	gRef Full	S/Y	28.71	28.80	28.77
MM-GDINO-T	gRef Full	Product	28.62	28.65	28.58
MM-GDINO-T	gRef Full	SIRC-style	28.87	28.86	28.85
MDETR-R101	gRef Full	Native	21.27	21.27	21.27
MDETR-R101	gRef Full	G/E	20.71	20.37	19.66
MDETR-R101	gRef Full	G/Y	19.70	20.49	20.07
MDETR-R101	gRef Full	S/E	20.46	20.68	20.42
MDETR-R101	gRef Full	S/Y	20.57	20.55	20.46
MDETR-R101	gRef Full	Product	20.41	20.05	19.50
MDETR-R101	gRef Full	SIRC-style	21.11	21.06	21.10

Table 15. Per-seed mixed-AUGRC effects, ×100: fixed seeds 17/42/73, their arithmetic mean and sample SD ($n - 1$ denominator), and the paired image-cluster 95% interval of the fixed three-seed mean effect. The image interval is not a training-seed interval: it can exclude zero even when the listed seed effects have different signs. No $n = 3$ t-test is used. Part 1.

Localizer	Population	Effect	Seed 17	Seed 42	Seed 73	Mean (SD)	Image CI
MM-GDINO-T	FineCops val	D_Y	-0.041	-0.021	-0.007	-0.023 (0.017)	[-0.048, +0.003]
MM-GDINO-T	FineCops val	D_E	-0.053	-0.041	-0.008	-0.034 (0.023)	[-0.068, +0.000]
MM-GDINO-T	FineCops val	I	+0.012	+0.020	+0.001	+0.011 (0.009)	[-0.016, +0.038]
MM-GDINO-T	FineCops val	$G : Y - E$	+0.182	+0.116	+0.148	+0.149 (0.033)	[+0.058, +0.235]
MM-GDINO-T	gRef source-disjoint	D_Y	-0.029	+0.017	+0.078	+0.022 (0.054)	[-0.013, +0.057]
MM-GDINO-T	gRef source-disjoint	D_E	-0.078	-0.012	+0.010	-0.027 (0.046)	[-0.060, +0.007]
MM-GDINO-T	gRef source-disjoint	I	+0.049	+0.030	+0.068	+0.049 (0.019)	[+0.013, +0.085]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	-0.001	-0.109	-0.089	-0.066 (0.057)	[-0.178, +0.043]
MM-GDINO-T	gRef Full	D_Y	-0.042	+0.005	+0.071	+0.011 (0.057)	[-0.021, +0.045]
MM-GDINO-T	gRef Full	D_E	-0.071	-0.010	+0.005	-0.025 (0.041)	[-0.055, +0.006]
MM-GDINO-T	gRef Full	I	+0.029	+0.015	+0.066	+0.037 (0.026)	[+0.003, +0.069]
MM-GDINO-T	gRef Full	$G : Y - E$	+0.031	-0.081	-0.070	-0.040 (0.062)	[-0.143, +0.062]

Table 16. Per-seed mixed-AUGRC effects, ×100: fixed seeds 17/42/73, their arithmetic mean and sample SD ($n-1$ denominator), and the paired image-cluster 95% interval of the fixed three-seed mean effect. The image interval is not a training-seed interval: it can exclude zero even when the listed seed effects have different signs. No $n=3$ t-test is used. Part 2.

Localizer	Population	Effect	Seed 17	Seed 42	Seed 73	Mean (SD)	Image CI
MDETR-R101	FineCops val	D_Y	-0.349	-0.075	-0.143	-0.189 (0.143)	[-0.257, -0.122]
MDETR-R101	FineCops val	D_E	-0.105	-0.226	-0.241	-0.191 (0.075)	[-0.285, -0.094]
MDETR-R101	FineCops val	I	-0.245	+0.151	+0.098	+0.001 (0.215)	[-0.091, +0.088]
MDETR-R101	FineCops val	$G:Y-E$	-0.664	-1.261	-0.672	-0.866 (0.342)	[-0.988, -0.745]
MDETR-R101	gRef source-disjoint	D_Y	+0.872	+0.068	+0.371	+0.437 (0.406)	[+0.363, +0.507]
MDETR-R101	gRef source-disjoint	D_E	-0.283	+0.283	+0.708	+0.236 (0.497)	[+0.137, +0.332]
MDETR-R101	gRef source-disjoint	I	+1.155	-0.215	-0.336	+0.201 (0.829)	[+0.108, +0.292]
MDETR-R101	gRef source-disjoint	$G:Y-E$	-1.032	+0.074	+0.392	-0.189 (0.748)	[-0.291, -0.083]
MDETR-R101	gRef Full	D_Y	+0.865	+0.059	+0.395	+0.440 (0.405)	[+0.373, +0.506]
MDETR-R101	gRef Full	D_E	-0.256	+0.305	+0.753	+0.268 (0.506)	[+0.176, +0.356]
MDETR-R101	gRef Full	I	+1.121	-0.247	-0.358	+0.172 (0.824)	[+0.085, +0.259]
MDETR-R101	gRef Full	$G:Y-E$	-1.008	+0.117	+0.405	-0.162 (0.747)	[-0.265, -0.065]

Table 17. All score endpoints, ×100: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 1.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MM-GDINO-T	FineCops val	Native	77.36 [76.18, 78.56]	45.36 [44.43, 46.28]	83.28 [82.22, 84.39]	96.09 [95.47, 96.62]	54.57 [53.75, 55.41]	28.11 [27.76, 28.46]
MM-GDINO-T	FineCops val	G/E	77.36 [76.18, 78.56]	62.62 [61.85, 63.39]	68.89 [67.49, 70.30]	80.11 [78.93, 81.21]	47.39 [46.52, 48.25]	26.23 [25.84, 26.60]
MM-GDINO-T	FineCops val	G/Y	77.36 [76.18, 78.56]	55.87 [54.97, 56.76]	80.55 [79.34, 81.74]	85.73 [84.83, 86.67]	47.69 [46.84, 48.52]	26.38 [26.01, 26.74]
MM-GDINO-T	FineCops val	S/E	77.36 [76.18, 78.56]	62.62 [61.83, 63.38]	69.22 [67.83, 70.61]	80.28 [79.15, 81.44]	47.26 [46.40, 48.12]	26.20 [25.81, 26.57]
MM-GDINO-T	FineCops val	S/Y	77.36 [76.18, 78.56]	56.21 [55.31, 57.09]	79.95 [78.74, 81.15]	87.15 [86.20, 88.01]	47.58 [46.74, 48.41]	26.36 [25.99, 26.71]
MM-GDINO-T	FineCops val	Product	77.36 [76.18, 78.56]	55.02 [54.17, 55.88]	80.94 [79.80, 82.04]	81.91 [80.91, 82.84]	48.90 [48.05, 49.73]	26.62 [26.25, 26.97]
MM-GDINO-T	FineCops val	SIRC-style	77.36 [76.18, 78.56]	50.89 [50.01, 51.76]	82.38 [81.30, 83.50]	82.15 [81.26, 83.05]	53.10 [52.26, 53.95]	27.35 [26.99, 27.71]
MDETR-R101	FineCops val	Native	65.56 [64.35, 66.73]	51.56 [50.76, 52.38]	74.62 [73.50, 75.75]	94.37 [93.64, 95.02]	57.59 [56.70, 58.48]	30.16 [29.77, 30.53]
MDETR-R101	FineCops val	G/E	65.56 [64.35, 66.73]	61.55 [60.84, 62.28]	64.90 [63.72, 66.06]	86.01 [85.12, 86.90]	55.23 [54.31, 56.14]	29.71 [29.30, 30.12]
MDETR-R101	FineCops val	G/Y	65.56 [64.35, 66.73]	56.83 [56.04, 57.64]	78.79 [77.79, 79.73]	90.40 [89.64, 91.12]	53.14 [52.28, 53.99]	28.84 [28.46, 29.22]
MDETR-R101	FineCops val	S/E	65.56 [64.35, 66.73]	61.49 [60.70, 62.26]	66.54 [65.31, 67.79]	86.50 [85.61, 87.43]	54.68 [53.71, 55.64]	29.52 [29.10, 29.92]
MDETR-R101	FineCops val	S/Y	65.56 [64.35, 66.73]	56.10 [55.27, 56.94]	81.17 [80.14, 82.16]	91.61 [90.85, 92.45]	53.10 [52.20, 53.95]	28.65 [28.27, 29.03]
MDETR-R101	FineCops val	Product	65.56 [64.35, 66.73]	60.52 [59.76, 61.26]	69.52 [68.40, 70.65]	87.87 [86.95, 88.80]	54.14 [53.23, 55.06]	29.33 [28.92, 29.73]
MDETR-R101	FineCops val	SIRC-style	65.56 [64.35, 66.73]	52.43 [51.64, 53.25]	74.83 [73.71, 75.95]	92.99 [92.26, 93.68]	57.22 [56.32, 58.12]	30.01 [29.62, 30.39]
MM-GDINO-T	gRef source-disjoint	Native	56.04 [54.42, 57.71]	59.55 [58.54, 60.55]	72.46 [70.87, 73.97]	81.36 [79.59, 83.13]	55.97 [54.66, 57.29]	30.05 [29.50, 30.57]
MM-GDINO-T	gRef source-disjoint	G/E	56.04 [54.42, 57.71]	75.79 [74.89, 76.67]	66.46 [64.78, 68.07]	72.89 [70.16, 75.73]	51.18 [49.90, 52.49]	28.68 [28.13, 29.21]
MM-GDINO-T	gRef source-disjoint	G/Y	56.04 [54.42, 57.71]	70.84 [69.93, 71.72]	72.99 [71.45, 74.51]	70.12 [67.49, 72.70]	50.79 [49.53, 52.07]	28.61 [28.07, 29.15]
MM-GDINO-T	gRef source-disjoint	S/E	56.04 [54.42, 57.71]	76.01 [75.09, 76.89]	66.58 [64.93, 68.18]	70.84 [67.71, 73.08]	51.04 [49.75, 52.34]	28.65 [28.10, 29.19]

Table 18. All score endpoints, ×100: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 2.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MM-GDINO-T	gRef source-disjoint	S/Y	56.04 [54.42, 57.71]	70.76 [69.82, 71.67]	72.64 [71.12, 74.13]	71.59 [69.01, 73.75]	50.72 [49.46, 52.00]	28.63 [28.09, 29.17]
MM-GDINO-T	gRef source-disjoint	Product	56.04 [54.42, 57.71]	72.46 [71.56, 73.34]	72.62 [71.02, 74.17]	72.17 [69.11, 75.11]	50.67 [49.41, 51.92]	28.48 [27.94, 29.01]
MM-GDINO-T	gRef source-disjoint	SIRC-style	56.04 [54.42, 57.71]	70.43 [69.50, 71.32]	73.43 [71.82, 74.99]	72.71 [69.87, 75.46]	52.43 [51.19, 53.71]	28.71 [28.18, 29.25]
MDETR-R101	gRef source-disjoint	Native	81.98 [80.32, 83.51]	65.35 [64.27, 66.37]	83.52 [81.94, 85.05]	88.32 [86.68, 90.56]	37.09 [36.14, 38.07]	21.13 [20.68, 21.59]
MDETR-R101	gRef source-disjoint	G/E	81.98 [80.32, 83.51]	72.35 [71.28, 73.36]	78.55 [76.54, 80.48]	82.13 [79.26, 85.97]	33.89 [32.97, 34.83]	20.11 [19.65, 20.57]
MDETR-R101	gRef source-disjoint	G/Y	81.98 [80.32, 83.51]	70.50 [69.36, 71.58]	84.75 [83.23, 86.22]	87.07 [84.44, 89.46]	32.68 [31.85, 33.55]	19.92 [19.48, 20.38]
MDETR-R101	gRef source-disjoint	S/E	81.98 [80.32, 83.51]	70.88 [69.76, 71.92]	78.75 [76.68, 80.73]	86.04 [83.22, 88.47]	34.24 [33.31, 35.19]	20.34 [19.89, 20.81]
MDETR-R101	gRef source-disjoint	S/Y	81.98 [80.32, 83.51]	68.60 [67.44, 69.71]	83.74 [82.08, 85.36]	88.76 [86.54, 90.80]	33.80 [32.90, 34.71]	20.35 [19.90, 20.82]
MDETR-R101	gRef source-disjoint	Product	81.98 [80.32, 83.51]	72.74 [71.63, 73.79]	80.69 [78.81, 82.50]	83.14 [79.66, 86.81]	32.88 [32.00, 33.79]	19.84 [19.39, 20.31]
MDETR-R101	gRef source-disjoint	SIRC-style	81.98 [80.32, 83.51]	66.01 [64.90, 67.05]	84.18 [82.58, 85.71]	87.25 [85.19, 89.40]	36.65 [35.71, 37.61]	20.95 [20.50, 21.40]
MM-GDINO-T	gRef Full	Native	55.95 [54.52, 57.47]	59.41 [58.47, 60.34]	71.79 [70.37, 73.11]	81.19 [79.65, 82.61]	56.38 [55.18, 57.55]	30.24 [29.76, 30.72]
MM-GDINO-T	gRef Full	G/E	55.95 [54.52, 57.47]	75.58 [74.75, 76.41]	66.52 [65.03, 67.97]	73.13 [70.61, 75.82]	51.35 [50.20, 52.52]	28.79 [28.31, 29.28]
MM-GDINO-T	gRef Full	G/Y	55.95 [54.52, 57.47]	70.75 [69.91, 71.56]	72.71 [71.32, 74.00]	70.18 [67.63, 72.50]	51.06 [49.90, 52.22]	28.75 [28.26, 29.23]
MM-GDINO-T	gRef Full	S/E	55.95 [54.52, 57.47]	75.75 [74.91, 76.61]	66.66 [65.18, 68.11]	71.11 [68.16, 73.32]	51.22 [50.08, 52.40]	28.76 [28.28, 29.25]
MM-GDINO-T	gRef Full	S/Y	55.95 [54.52, 57.47]	70.70 [69.82, 71.53]	72.43 [71.07, 73.72]	71.68 [69.43, 73.67]	50.96 [49.80, 52.12]	28.76 [28.28, 29.25]
MM-GDINO-T	gRef Full	Product	55.95 [54.52, 57.47]	72.25 [71.40, 73.08]	72.51 [71.09, 73.84]	72.25 [69.35, 74.99]	50.94 [49.79, 52.08]	28.62 [28.14, 29.10]
MM-GDINO-T	gRef Full	SIRC-style	55.95 [54.52, 57.47]	70.18 [69.35, 71.02]	73.17 [71.71, 74.51]	72.82 [70.11, 75.59]	52.75 [51.61, 53.86]	28.86 [28.38, 29.34]
MDETR-R101	gRef Full	Native	81.84 [80.41, 83.31]	65.22 [64.30, 66.18]	83.41 [81.95, 84.82]	88.30 [86.69, 90.24]	37.43 [36.59, 38.31]	21.27 [20.86, 21.69]

Table 19. All score endpoints, ×100: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 3.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MDETR-R101	gRef Full	G/E	81.84 [80.41, 83.31]	72.28 [71.34, 73.26]	78.32 [76.49, 80.13]	81.64 [79.12, 84.62]	34.22 [33.39, 35.08]	20.25 [19.84, 20.67]
MDETR-R101	gRef Full	G/Y	81.84 [80.41, 83.31]	70.49 [69.50, 71.51]	84.07 [82.47, 85.58]	86.17 [83.95, 88.46]	33.10 [32.33, 33.91]	20.09 [19.69, 20.52]
MDETR-R101	gRef Full	S/E	81.84 [80.41, 83.31]	70.68 [69.70, 71.70]	78.44 [76.49, 80.31]	85.80 [83.30, 88.05]	34.70 [33.83, 35.59]	20.52 [20.10, 20.94]
MDETR-R101	gRef Full	S/Y	81.84 [80.41, 83.31]	68.57 [67.55, 69.63]	83.08 [81.42, 84.66]	88.58 [86.55, 90.38]	34.20 [33.36, 35.06]	20.53 [20.12, 20.95]
MDETR-R101	gRef Full	Product	81.84 [80.41, 83.31]	72.65 [71.69, 73.66]	80.47 [78.76, 82.16]	82.43 [79.37, 85.61]	33.25 [32.44, 34.10]	19.99 [19.58, 20.42]
MDETR-R101	gRef Full	SIRC-style	81.84 [80.41, 83.31]	65.90 [64.95, 66.88]	84.05 [82.58, 85.48]	87.01 [85.04, 88.93]	37.00 [36.15, 37.87]	21.09 [20.68, 21.50]

Table 20. FineCops val error composition at requested coverage 50/90%. Achieved coverage is the equal-seed mean fraction of all requests retained, in percent; the boundary tie group is accepted in full. W, N and mixed are conditional error probabilities among accepted requests, $\times 100$, with paired image-cluster 95% intervals. They are not counts or failure rates divided by all requests. W+N=mixed before display rounding. No deployment threshold is fitted; all other coverages/surfaces remain in the full analysis JSON.

Localizer	Head	Requested (%)	Achieved (%)	W|accept	N|accept	Failure|accept
MM-GDINO-T	G/E	50	50.003	8.44 [7.67, 9.25]	41.46 [40.45, 42.40]	49.90 [48.97, 50.76]
MM-GDINO-T	G/E	90	90.003	12.00 [11.25, 12.77]	45.26 [44.33, 46.13]	57.25 [56.50, 57.95]
MM-GDINO-T	G/Y	50	50.003	3.55 [3.13, 4.03]	46.81 [45.95, 47.65]	50.36 [49.53, 51.23]
MM-GDINO-T	G/Y	90	90.003	11.04 [10.34, 11.80]	46.35 [45.45, 47.20]	57.39 [56.68, 58.08]
MM-GDINO-T	S/E	50	50.003	8.41 [7.67, 9.19]	41.45 [40.47, 42.39]	49.87 [48.92, 50.73]
MM-GDINO-T	S/E	90	90.003	11.98 [11.23, 12.76]	45.25 [44.34, 46.13]	57.23 [56.50, 57.93]
MM-GDINO-T	S/Y	50	50.003	3.83 [3.38, 4.36]	46.45 [45.56, 47.28]	50.28 [49.43, 51.14]
MM-GDINO-T	S/Y	90	90.003	10.85 [10.13, 11.58]	46.69 [45.80, 47.53]	57.54 [56.81, 58.21]
MDETR-R101	G/E	50	50.003	15.38 [14.52, 16.25]	41.06 [40.09, 42.02]	56.44 [55.49, 57.40]
MDETR-R101	G/E	90	90.003	17.74 [16.97, 18.55]	46.35 [45.46, 47.20]	64.09 [63.35, 64.81]
MDETR-R101	G/Y	50	50.003	9.04 [8.38, 9.78]	44.52 [43.55, 45.43]	53.56 [52.67, 54.44]
MDETR-R101	G/Y	90	90.003	16.29 [15.54, 17.07]	47.34 [46.49, 48.19]	63.63 [62.89, 64.37]
MDETR-R101	S/E	50	50.003	14.76 [13.86, 15.66]	40.98 [39.94, 41.99]	55.74 [54.74, 56.69]
MDETR-R101	S/E	90	90.003	17.52 [16.74, 18.32]	46.45 [45.58, 47.31]	63.97 [63.22, 64.69]
MDETR-R101	S/Y	50	50.003	8.02 [7.34, 8.72]	44.91 [43.96, 45.88]	52.94 [52.02, 53.86]
MDETR-R101	S/Y	90	90.003	15.78 [15.02, 16.56]	47.68 [46.81, 48.50]	63.46 [62.71, 64.17]

Table 21. Frozen weights, alternate inference readout: mixed AUGRC $\times 100$. Matched and alternate values are seed mean (sample SD); differences use paired 95% intervals. Every row keeps Native boxes fixed. These off-diagonal readings diagnose the training/deployment readout contract; they are not substituted for the matched four-cell matrix. Part 1.

Localizer	Population	Trained head	Alternate read	Matched (SD)	Alternate (SD)	Alt.−matched [CI]
MM-GDINO-T	FineCops val	G/Y	S	26.38 (0.04)	26.44 (0.03)	+0.058 [+0.019, +0.097]
MM-GDINO-T	FineCops val	G/E	S	26.23 (0.07)	26.25 (0.05)	+0.016 [-0.030, +0.061]
MM-GDINO-T	FineCops val	S/Y	G	26.36 (0.04)	28.38 (0.17)	+2.029 [+1.876, +2.179]
MM-GDINO-T	FineCops val	S/E	G	26.20 (0.07)	27.70 (0.49)	+1.506 [+1.365, +1.645]
MDETR-R101	FineCops val	G/Y	S	28.84 (0.13)	30.45 (2.83)	+1.608 [+1.513, +1.698]
MDETR-R101	FineCops val	G/E	S	29.71 (0.27)	32.51 (2.87)	+2.799 [+2.642, +2.952]
MDETR-R101	FineCops val	S/Y	G	28.65 (0.02)	29.09 (0.08)	+0.435 [+0.371, +0.495]
MDETR-R101	FineCops val	S/E	G	29.52 (0.28)	29.97 (0.24)	+0.449 [+0.361, +0.532]
MM-GDINO-T	gRef source-disjoint	G/Y	S	28.61 (0.04)	28.62 (0.04)	+0.012 [-0.010, +0.036]
MM-GDINO-T	gRef source-disjoint	G/E	S	28.68 (0.08)	28.57 (0.08)	-0.105 [-0.139, -0.069]
MM-GDINO-T	gRef source-disjoint	S/Y	G	28.63 (0.05)	30.50 (0.23)	+1.866 [+1.672, +2.059]
MM-GDINO-T	gRef source-disjoint	S/E	G	28.65 (0.11)	29.91 (0.30)	+1.260 [+1.117, +1.400]
MDETR-R101	gRef source-disjoint	G/Y	S	19.92 (0.38)	22.77 (3.43)	+2.853 [+2.740, +2.963]
MDETR-R101	gRef source-disjoint	G/E	S	20.11 (0.54)	24.47 (3.53)	+4.362 [+4.173, +4.542]
MDETR-R101	gRef source-disjoint	S/Y	G	20.35 (0.06)	20.25 (0.11)	-0.109 [-0.167, -0.049]
MDETR-R101	gRef source-disjoint	S/E	G	20.34 (0.15)	20.40 (0.01)	+0.057 [-0.010, +0.128]
MM-GDINO-T	gRef Full	G/Y	S	28.75 (0.05)	28.77 (0.05)	+0.017 [-0.006, +0.040]
MM-GDINO-T	gRef Full	G/E	S	28.79 (0.08)	28.70 (0.09)	-0.095 [-0.126, -0.063]

Table 22. Frozen weights, alternate inference readout: mixed AUGRC $\times 100$. Matched and alternate values are seed mean (sample SD); differences use paired 95% intervals. Every row keeps Native boxes fixed. These off-diagonal readings diagnose the training/deployment readout contract; they are not substituted for the matched four-cell matrix. Part 2.

Localizer	Population	Trained head	Alternate read	Matched (SD)	Alternate (SD)	Alt.−matched [CI]
MM-GDINO-T	gRef Full	S/Y	G	28.76 (0.05)	30.61 (0.22)	+1.851 [+1.680, +2.024]
MM-GDINO-T	gRef Full	S/E	G	28.76 (0.11)	30.02 (0.29)	+1.254 [+1.125, +1.377]
MDETR-R101	gRef Full	G/Y	S	20.09 (0.39)	22.92 (3.37)	+2.834 [+2.735, +2.935]
MDETR-R101	gRef Full	G/E	S	20.25 (0.53)	24.62 (3.52)	+4.369 [+4.202, +4.533]
MDETR-R101	gRef Full	S/Y	G	20.53 (0.06)	20.42 (0.11)	-0.105 [-0.159, -0.053]
MDETR-R101	gRef Full	S/E	G	20.52 (0.14)	20.55 (0.03)	+0.034 [-0.027, +0.097]

Table 23. Same-image state comparisons: exact participating images, requests and within-image pairs. The conditional and comparable-unconditional estimates use these same participating requests; bootstrap image multiplicity applies once to each within-copy pair.

Localizer	Population	State pair	Images	Requests	Within-image pairs
MM-GDINO-T	FineCops val	CW	835	3943	5016
MM-GDINO-T	FineCops val	CN	2344	14749	32776
MM-GDINO-T	FineCops val	WN	624	4040	6110
MDETR-R101	FineCops val	CW	1177	5340	6921
MDETR-R101	FineCops val	CN	2194	13350	26598
MDETR-R101	FineCops val	WN	1088	7322	12288
MM-GDINO-T	gRef source-disjoint	CW	1044	8322	15552
MM-GDINO-T	gRef source-disjoint	CN	1215	12871	33120
MM-GDINO-T	gRef source-disjoint	WN	1103	10979	25726
MDETR-R101	gRef source-disjoint	CW	584	5198	8744
MDETR-R101	gRef source-disjoint	CN	1266	15723	48801
MDETR-R101	gRef source-disjoint	WN	593	5326	10045
MM-GDINO-T	gRef Full	CW	1232	9808	18448
MM-GDINO-T	gRef Full	CN	1430	15171	39160
MM-GDINO-T	gRef Full	WN	1299	12973	30428
MDETR-R101	gRef Full	CW	692	6166	10397
MDETR-R101	gRef Full	CN	1488	18517	57647
MDETR-R101	gRef Full	WN	702	6351	11941

Table 24. FineCops original-positive-difficulty counts. High/low states are the first/second state in the pair label. Zero low-state counts at L2/L3 expose the absence of same-level no-target comparisons; these cells are not assigned an AUROC. Images overlap across levels/state comparisons.

Localizer	State pair	Level	High-state requests	Low-state requests	Images
MM-GDINO-T	CW	L1	5506	591	2710
MM-GDINO-T	CW	L2	1566	1318	1332
MM-GDINO-T	CW	L3	220	225	269
MM-GDINO-T	CN	L1	5506	9029	2686
MM-GDINO-T	CN	L2	1566	0	966
MM-GDINO-T	CN	L3	220	0	168
MM-GDINO-T	WN	L1	591	9029	2458
MM-GDINO-T	WN	L2	1318	0	805
MM-GDINO-T	WN	L3	225	0	167
MDETR-R101	CW	L1	4581	1516	2710
MDETR-R101	CW	L2	1393	1491	1332
MDETR-R101	CW	L3	206	239	269
MDETR-R101	CN	L1	4581	9029	2673
MDETR-R101	CN	L2	1393	0	909
MDETR-R101	CN	L3	206	0	157
MDETR-R101	WN	L1	1516	9029	2476
MDETR-R101	WN	L2	1491	0	889
MDETR-R101	WN	L3	239	0	172

Table 25. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S - G$ is the emission readout effect; $G : Y - E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 1.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MM-GDINO-T	FineCops val	$Y : S - G$	Image CW	-0.405 [-0.822, +0.012]	+0.159 [-0.661, +0.916]	-0.565 [-1.219, +0.148]
MM-GDINO-T	FineCops val	$Y : S - G$	Level CW	-0.600 [-0.829, -0.372]	-0.591 [-0.889, -0.299]	-0.009 [-0.156, +0.138]
MM-GDINO-T	FineCops val	$G : Y - E$	Image CW	+8.854 [+7.523, +10.221]	+7.642 [+5.850, +9.421]	+1.211 [-0.222, +2.634]
MM-GDINO-T	FineCops val	$G : Y - E$	Level CW	+11.654 [+10.721, +12.641]	+6.940 [+5.993, +7.944]	+4.713 [+4.075, +5.363]
MM-GDINO-T	FineCops val	$Y : S - G$	Image CN	+0.241 [+0.129, +0.349]	+0.289 [+0.066, +0.512]	-0.048 [-0.231, +0.139]
MM-GDINO-T	FineCops val	$Y : S - G$	Level CN	+0.261 [+0.154, +0.366]	+0.216 [+0.112, +0.317]	+0.045 [-0.020, +0.109]
MM-GDINO-T	FineCops val	$G : Y - E$	Image CN	-1.700 [-2.002, -1.400]	-1.163 [-1.557, -0.760]	-0.537 [-0.877, -0.202]
MM-GDINO-T	FineCops val	$G : Y - E$	Level CN	-3.523 [-3.884, -3.166]	+0.224 [+0.024, +0.431]	-3.747 [-4.076, -3.434]
MM-GDINO-T	FineCops val	$Y : S - G$	Image WN	+0.397 [-0.016, +0.792]	+0.660 [-0.026, +1.326]	-0.263 [-0.854, +0.331]
MM-GDINO-T	FineCops val	$Y : S - G$	Level WN	+0.619 [+0.366, +0.863]	+0.220 [-0.117, +0.540]	+0.399 [+0.085, +0.719]
MM-GDINO-T	FineCops val	$G : Y - E$	Image WN	-12.753 [-14.160, -11.363]	-10.644 [-12.590, -8.700]	-2.109 [-3.612, -0.580]
MM-GDINO-T	FineCops val	$G : Y - E$	Level WN	-17.795 [-18.821, -16.781]	-8.720 [-9.994, -7.505]	-9.075 [-10.290, -7.813]
MDETR-R101	FineCops val	$Y : S - G$	Image CW	+2.986 [+2.370, +3.600]	+2.712 [+1.668, +3.786]	+0.275 [-0.529, +1.106]
MDETR-R101	FineCops val	$Y : S - G$	Level CW	+2.373 [+1.929, +2.839]	+2.586 [+2.093, +3.107]	-0.213 [-0.434, +0.000]
MDETR-R101	FineCops val	$G : Y - E$	Image CW	+12.180 [+11.106, +13.269]	+11.752 [+10.280, +13.309]	+0.429 [-0.665, +1.493]
MDETR-R101	FineCops val	$G : Y - E$	Level CW	+13.889 [+13.044, +14.743]	+13.660 [+12.738, +14.596]	+0.229 [-0.192, +0.661]
MDETR-R101	FineCops val	$Y : S - G$	Image CN	+0.727 [+0.392, +1.066]	+0.924 [+0.410, +1.456]	-0.197 [-0.587, +0.206]
MDETR-R101	FineCops val	$Y : S - G$	Level CN	+0.301 [-0.016, +0.622]	+0.834 [+0.514, +1.166]	-0.533 [-0.719, -0.349]

Table 26. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S - G$ is the emission readout effect; $G : Y - E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 2.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MDETR-R101	FineCops val	$G : Y - E$	Image CN	+0.450 [-0.121, +1.027]	+1.307 [+0.509, +2.098]	-0.857 [-1.453, -0.265]
MDETR-R101	FineCops val	$G : Y - E$	Level CN	+0.290 [-0.265, +0.857]	+2.195 [+1.647, +2.750]	-1.905 [-2.213, -1.595]
MDETR-R101	FineCops val	$Y : S - G$	Image WN	-3.433 [-4.085, -2.798]	-2.772 [-3.850, -1.762]	-0.661 [-1.466, +0.165]
MDETR-R101	FineCops val	$Y : S - G$	Level WN	-2.696 [-3.193, -2.211]	-3.157 [-3.834, -2.501]	+0.461 [+0.001, +0.954]
MDETR-R101	FineCops val	$G : Y - E$	Image WN	-10.839 [-11.825, -9.875]	-9.608 [-11.090, -8.055]	-1.230 [-2.362, -0.072]
MDETR-R101	FineCops val	$G : Y - E$	Level WN	-14.257 [-15.124, -13.406]	-13.524 [-14.635, -12.448]	-0.733 [-1.527, +0.055]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image CW	-0.378 [-0.672, -0.095]	-0.039 [-0.829, +0.750]	-0.340 [-1.047, +0.387]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image CW	+6.070 [+5.150, +6.975]	+3.920 [+2.668, +5.218]	+2.150 [+1.000, +3.341]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image CN	+0.036 [-0.114, +0.185]	+0.430 [+0.164, +0.701]	-0.394 [-0.607, -0.179]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image CN	-3.316 [-3.764, -2.885]	-4.266 [-4.832, -3.717]	+0.950 [+0.500, +1.404]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image WN	-0.258 [-0.529, +0.003]	-0.277 [-0.767, +0.196]	+0.019 [-0.384, +0.408]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image WN	-6.775 [-7.481, -6.061]	-5.456 [-6.318, -4.548]	-1.319 [-2.107, -0.578]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image CW	-1.301 [-2.039, -0.532]	-1.616 [-2.740, -0.541]	+0.315 [-0.616, +1.282]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image CW	+4.478 [+3.273, +5.734]	+7.087 [+5.545, +8.782]	-2.608 [-3.920, -1.291]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image CN	-1.927 [-2.211, -1.634]	-1.540 [-1.914, -1.175]	-0.386 [-0.664, -0.111]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image CN	-0.509 [-0.934, -0.098]	+1.155 [+0.680, +1.617]	-1.664 [-2.050, -1.278]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image WN	-2.145 [-3.106, -1.217]	-2.150 [-3.233, -1.075]	+0.005 [-0.804, +0.855]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image WN	-5.580 [-7.305, -4.013]	-5.044 [-6.793, -3.358]	-0.536 [-1.760, +0.681]

Table 27. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S - G$ is the emission readout effect; $G : Y - E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 3.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MM-GDINO-T	gRef Full	$Y : S - G$	Image CW	-0.286 [-0.559, -0.014]	-0.069 [-0.783, +0.641]	-0.217 [-0.846, +0.447]
MM-GDINO-T	gRef Full	$G : Y - E$	Image CW	+5.685 [+4.847, +6.520]	+3.905 [+2.740, +5.049]	+1.780 [+0.701, +2.839]
MM-GDINO-T	gRef Full	$Y : S - G$	Image CN	+0.077 [-0.067, +0.211]	+0.375 [+0.116, +0.629]	-0.299 [-0.500, -0.094]
MM-GDINO-T	gRef Full	$G : Y - E$	Image CN	-3.276 [-3.676, -2.861]	-4.199 [-4.727, -3.663]	+0.923 [+0.523, +1.342]
MM-GDINO-T	gRef Full	$Y : S - G$	Image WN	-0.244 [-0.485, +0.000]	-0.348 [-0.792, +0.085]	+0.104 [-0.250, +0.453]
MM-GDINO-T	gRef Full	$G : Y - E$	Image WN	-6.524 [-7.173, -5.869]	-5.531 [-6.309, -4.721]	-0.993 [-1.692, -0.298]
MDETR-R101	gRef Full	$Y : S - G$	Image CW	-1.217 [-1.916, -0.546]	-1.552 [-2.567, -0.611]	+0.335 [-0.450, +1.193]
MDETR-R101	gRef Full	$G : Y - E$	Image CW	+4.117 [+2.833, +5.388]	+6.620 [+5.146, +8.235]	-2.503 [-3.693, -1.282]
MDETR-R101	gRef Full	$Y : S - G$	Image CN	-1.946 [-2.205, -1.669]	-1.689 [-2.013, -1.342]	-0.257 [-0.508, -0.009]
MDETR-R101	gRef Full	$G : Y - E$	Image CN	-0.534 [-0.914, -0.138]	+1.051 [+0.593, +1.497]	-1.585 [-1.933, -1.227]
MDETR-R101	gRef Full	$Y : S - G$	Image WN	-2.245 [-3.087, -1.451]	-2.116 [-3.190, -1.050]	-0.129 [-0.975, +0.733]
MDETR-R101	gRef Full	$G : Y - E$	Image WN	-5.285 [-6.869, -3.778]	-4.958 [-6.420, -3.450]	-0.328 [-1.447, +0.778]

Table 28. True edited-negative/positive-parent pair wins on FineCops, $\times 100$. Raw columns are equal-seed mean pair wins; effects include paired image-cluster 95% intervals. C/W refers to the parent's Native correctness. Pair counts repeat positive parents; C/W image counts may overlap. These are diagnostic pair wins, not official Recall@1.

Localizer	Parent	Pairs (images)	G/E	G/Y	S/Y	$G : Y - E$	$Y : S - G$
MM-GDINO-T	all	9029 (2433)	67.09	66.60	66.42	-0.495 [-0.986, -0.027]	-0.174 [-0.563, +0.196]
MM-GDINO-T	C	8136 (2303)	67.99	67.76	67.52	-0.229 [-0.651, +0.208]	-0.234 [-0.606, +0.134]
MM-GDINO-T	W	893 (348)	58.94	56.03	56.40	-2.912 [-5.905, +0.000]	+0.373 [-1.407, +2.114]
MDETR-R101	all	9029 (2433)	64.52	62.83	62.96	-1.683 [-2.562, -0.823]	+0.126 [-0.557, +0.811]
MDETR-R101	C	6675 (2092)	66.78	67.92	69.20	+1.144 [+0.187, +2.068]	+1.283 [+0.548, +2.026]
MDETR-R101	W	2354 (845)	58.11	48.41	45.26	-9.700 [-11.789, -7.615]	-3.158 [-4.789, -1.644]

Table 29. Dense-logit global-winner geometry, $\times 100$; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 1.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	FineCops val	G/E	C	7292	7.94	93.94	–
MM-GDINO-T	FineCops val	G/E	W	2134	29.94	72.01	14.18
MM-GDINO-T	FineCops val	G/E	N	9029	10.40	92.79	–
MM-GDINO-T	FineCops val	G/Y	C	7292	4.85	96.71	–
MM-GDINO-T	FineCops val	G/Y	W	2134	18.20	83.25	8.83
MM-GDINO-T	FineCops val	G/Y	N	9029	6.65	96.13	–
MM-GDINO-T	FineCops val	S/E	C	7292	48.04	84.23	–
MM-GDINO-T	FineCops val	S/E	W	2134	66.67	55.43	16.59
MM-GDINO-T	FineCops val	S/E	N	9029	66.60	78.92	–
MM-GDINO-T	FineCops val	S/Y	C	7292	59.94	86.40	–
MM-GDINO-T	FineCops val	S/Y	W	2134	84.88	60.79	17.37
MM-GDINO-T	FineCops val	S/Y	N	9029	72.06	82.22	–
MDETR-R101	FineCops val	G/E	C	6180	72.21	42.33	–
MDETR-R101	FineCops val	G/E	W	3246	81.91	31.42	8.93
MDETR-R101	FineCops val	G/E	N	9029	77.17	36.64	–
MDETR-R101	FineCops val	G/Y	C	6180	58.14	63.87	–
MDETR-R101	FineCops val	G/Y	W	3246	62.62	53.96	8.88
MDETR-R101	FineCops val	G/Y	N	9029	58.45	60.37	–
MDETR-R101	FineCops val	S/E	C	6180	41.49	69.26	–
MDETR-R101	FineCops val	S/E	W	3246	65.10	48.61	17.74
MDETR-R101	FineCops val	S/E	N	9029	57.25	56.50	–
MDETR-R101	FineCops val	S/Y	C	6180	15.72	89.32	–
MDETR-R101	FineCops val	S/Y	W	3246	54.64	55.69	19.42
MDETR-R101	FineCops val	S/Y	N	9029	32.62	75.15	–

Table 30. Dense-logit global-winner geometry, $\times 100$; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 2.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	gRef source-disjoint	G/E	C	5519	15.14	86.57	–
MM-GDINO-T	gRef source-disjoint	G/E	W	4329	33.65	69.90	10.98
MM-GDINO-T	gRef source-disjoint	G/E	N	7716	34.37	70.80	–
MM-GDINO-T	gRef source-disjoint	G/Y	C	5519	8.61	92.61	–
MM-GDINO-T	gRef source-disjoint	G/Y	W	4329	22.58	80.74	8.39
MM-GDINO-T	gRef source-disjoint	G/Y	N	7716	23.52	81.38	–
MM-GDINO-T	gRef source-disjoint	S/E	C	5519	55.92	63.45	–
MM-GDINO-T	gRef source-disjoint	S/E	W	4329	77.99	43.94	12.10
MM-GDINO-T	gRef source-disjoint	S/E	N	7716	88.02	40.75	–
MM-GDINO-T	gRef source-disjoint	S/Y	C	5519	70.22	70.72	–
MM-GDINO-T	gRef source-disjoint	S/Y	W	4329	92.22	52.96	16.01
MM-GDINO-T	gRef source-disjoint	S/Y	N	7716	92.12	52.06	–
MDETR-R101	gRef source-disjoint	G/E	C	8073	71.09	46.48	–
MDETR-R101	gRef source-disjoint	G/E	W	1775	80.11	34.12	10.35
MDETR-R101	gRef source-disjoint	G/E	N	7716	77.06	38.10	–
MDETR-R101	gRef source-disjoint	G/Y	C	8073	66.26	63.31	–
MDETR-R101	gRef source-disjoint	G/Y	W	1775	62.25	57.50	8.30
MDETR-R101	gRef source-disjoint	G/Y	N	7716	63.57	60.09	–
MDETR-R101	gRef source-disjoint	S/E	C	8073	32.08	78.03	–
MDETR-R101	gRef source-disjoint	S/E	W	1775	64.09	50.90	17.71
MDETR-R101	gRef source-disjoint	S/E	N	7716	56.70	56.49	–
MDETR-R101	gRef source-disjoint	S/Y	C	8073	19.08	87.00	–
MDETR-R101	gRef source-disjoint	S/Y	W	1775	51.55	59.35	11.44
MDETR-R101	gRef source-disjoint	S/Y	N	7716	37.61	71.66	–

Table 31. Dense-logit global-winner geometry, ×100; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 3.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	gRef Full	G/E	C	6469	15.22	86.46	–
MM-GDINO-T	gRef Full	G/E	W	5094	33.08	70.46	10.89
MM-GDINO-T	gRef Full	G/E	N	9121	34.13	71.00	–
MM-GDINO-T	gRef Full	G/Y	C	6469	8.69	92.49	–
MM-GDINO-T	gRef Full	G/Y	W	5094	22.01	81.18	8.24
MM-GDINO-T	gRef Full	G/Y	N	9121	23.31	81.47	–
MM-GDINO-T	gRef Full	S/E	C	6469	55.72	63.43	–
MM-GDINO-T	gRef Full	S/E	W	5094	77.76	44.13	12.05
MM-GDINO-T	gRef Full	S/E	N	9121	87.64	40.92	–
MM-GDINO-T	gRef Full	S/Y	C	6469	70.06	70.48	–
MM-GDINO-T	gRef Full	S/Y	W	5094	92.08	52.91	15.61
MM-GDINO-T	gRef Full	S/Y	N	9121	92.02	52.16	–
MDETR-R101	gRef Full	G/E	C	9463	71.11	46.38	–
MDETR-R101	gRef Full	G/E	W	2100	80.03	34.32	10.29
MDETR-R101	gRef Full	G/E	N	9121	76.98	38.19	–
MDETR-R101	gRef Full	G/Y	C	9463	66.28	63.29	–
MDETR-R101	gRef Full	G/Y	W	2100	62.37	57.42	8.30
MDETR-R101	gRef Full	G/Y	N	9121	63.59	59.98	–
MDETR-R101	gRef Full	S/E	C	9463	32.31	77.86	–
MDETR-R101	gRef Full	S/E	W	2100	62.90	51.92	17.37
MDETR-R101	gRef Full	S/E	N	9121	56.45	56.56	–
MDETR-R101	gRef Full	S/Y	C	9463	19.05	87.02	–
MDETR-R101	gRef Full	S/Y	W	2100	50.24	60.50	11.48
MDETR-R101	gRef Full	S/Y	N	9121	37.60	71.56	–

Table 32. AUGRC crossover prior (percent). Interior-only intervals are conditional summaries; non-interior replicates remain counted out of 5,000. Part 1.

Localizer	Population	Contrast	Prior	Root status	All-draw CI	Interior-only CI	Interior draws
MM-GDINO-T	FineCops val	global_emit_minus_exists	42.82	interior	[39.32, 46.40]	[39.32, 46.40]	5000
MM-GDINO-T	FineCops val	selected_emit_minus_exists	41.95	interior	[38.57, 45.56]	[38.57, 45.56]	5000
MM-GDINO-T	FineCops val	D_emit	34.23	interior	[21.42, 51.00]	[21.42, 51.00]	5000
MDETR-R101	FineCops val	global_emit_minus_exists	–	no interior root	–	[91.11, 99.88]	759
MDETR-R101	FineCops val	selected_emit_minus_exists	–	no interior root	–	[88.00, 99.80]	2372
MDETR-R101	FineCops val	D_emit	–	no interior root	–	[76.87, 99.86]	157
MM-GDINO-T	gRef source-disjoint	global_emit_minus_exists	47.42	interior	[41.70, 53.26]	[41.70, 53.26]	5000
MM-GDINO-T	gRef source-disjoint	selected_emit_minus_exists	44.87	interior	[39.16, 50.57]	[39.16, 50.57]	5000
MM-GDINO-T	gRef source-disjoint	D_emit	80.20	interior	–	[22.13, 97.92]	3419
MDETR-R101	gRef source-disjoint	global_emit_minus_exists	69.42	interior	–	[53.22, 91.37]	4955
MDETR-R101	gRef source-disjoint	selected_emit_minus_exists	42.63	interior	[31.18, 58.77]	[31.18, 58.77]	5000
MDETR-R101	gRef source-disjoint	D_emit	–	no interior root	–	[1.28, 1.28]	1
MM-GDINO-T	gRef Full	global_emit_minus_exists	46.28	interior	[40.79, 51.82]	[40.79, 51.82]	5000
MM-GDINO-T	gRef Full	selected_emit_minus_exists	44.30	interior	[38.88, 49.77]	[38.88, 49.77]	5000
MM-GDINO-T	gRef Full	D_emit	61.97	interior	–	[13.87, 96.12]	4282
MDETR-R101	gRef Full	global_emit_minus_exists	66.71	interior	–	[51.55, 88.32]	4973
MDETR-R101	gRef Full	selected_emit_minus_exists	42.99	interior	[31.46, 58.63]	[31.46, 58.63]	5000
MDETR-R101	gRef Full	D_emit	–	no interior root	–	–	0

H. Reusable Evaluation

The released record interface stores sample/image identities, fixed Native correctness and the compared score slots. It supports state-pair AUROCs, complete-risk integrals, conditional comparisons and paired image resampling without rerunning a detector. Versioned source bindings keep the two studies and their frozen references distinct. Synthetic tests cover score ties, no-target semantics, exact risk/interaction identities, negative interactions without absolute Y gains, complete positive sampling, frozen ownership and replicate-wise q05. This is an implementation of the empirical study rather than an additional methodological novelty claim.

References

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